Renewable Energy 83 (2015) 305–317

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

MPPT strategy based on speed control for AWS-based wave energy conversion system

Mostafa I. Marei*, Mohamed Mokhtar, Ahmed A. El-Sattar

Department of Electrical Power and Machines, Faculty of Engineering, Ain Shams University, Cairo 11517, Egypt

ARTICLE INFO

Article history:
Received 29 March 2014
Accepted 16 April 2015
Available online 14 May 2015

Keywords:
Archimedes wave swing
Estimation
Low voltage ride-through
Maximum power point tracking
Wave energy conversion system

ABSTRACT

One of the attractive direct-drive wave energy conversion systems is the Archimedes Wave Swing (AWS) coupled to a Linear Permanent Magnet Synchronous Generator (LPMSG). This paper presents an integrated control strategy for the back-to-back converter interfacing the LPMSG not only to extract the maximum power from the wave, but also to ride-through the fault. The proposed maximum power tracking technique is based on speed sensorless control of the LPMSG. The unscented Kalman filter is adapted to estimate the translator velocity. The optimal velocity is obtained from the instantaneous active power at the generator terminals. Moreover, a low-voltage ride-through control is integrated to satisfy the grid-code requirements by injecting reactive current during grid disturbances. The generated active power at the fault instant is considered in determining the dynamic reactive power injection to not exceed the ratings of the grid-side converter. The superiority of the proposed strategy is the result of its ability to regulate the translator velocity that generates optimum power. Numerical simulations are conducted to evaluate the dynamic performance of the proposed integrated optimal strategy. Besides, it has been shown that the proposed methodology outdoes others by the decreased power fluctuations which leads to a reduction of the converter size.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

Recently, Wave Energy Conversion Systems (WECS) gain an increasing attention. There are many methods for capturing energy from waves and convert it to electrical energy [1]. Mechanical interfaces such as Oscillating Water Column (OWC) [2], water turbines [3], and hydraulic systems [4] are used to convert the linear motion to rotational and hence conventional rotary electrical machines such as induction or synchronous generators can be used to generate electricity. However, the main drawbacks of these WECS are their mechanical complexity and low energy-conversion efficiency. Alternatively, direct-drive WECS, which are based on a linear generator connected directly to a buoy, eliminate the need for complex mechanical interface. As a result of reducing the moving parts, the overall system efficiency is increased and the maintenance rate is decreased. Many linear generators can be coupled with the direct-drive WECS such as induction generator with or without iron in the translator, switched reluctance generator, transverse flux machine, vernier hybrid machine, and Linear Permanent Magnet Synchronous Generator (LPMSG) [5–7].

Archimedes Wave Swing (AWS) is the first wave energy conversion device to adopt the direct-drive power take off. As shown in Fig. 1, the AWS consists of two parts: hollow cylinder and its cap which is called the floater. The hollow cylinder is fixed at the seabed and filled with air while the floater oscillates vertically by waves. When a wave is above the AWS, the floater moves down due to the weight of the water above the AWS which pressurizes the air inside the device. In contrast, the weight of the water decreases when a wave trough is above the AWS. As a result, the floater moves up by the expanded air volume inside the cylinder [8,9]. The LPMSG is connected directly to the AWS to convert the linear motion of the floater to electrical energy. As the excitation force of the wave changes, the output from the generator varies in magnitude and frequency. Back-to-back converters are used widely to deliver the oscillatory power from the LPMSG to the grid.

Many control methods are proposed for the back-to-back converters to extract the maximum power from the wave [10–14]. Regulating the LPMSG based AWS to achieve mechanical resonance is presented in Ref. [10]. Although this method is simple, it requires sensors to measure the wave speed, magnitude and direction, as

* Corresponding author. Tel.: +20 (02) 0100 6957052.
E-mail address: mostafamarei@yahoo.ca (M.I. Marei).

http://dx.doi.org/10.1016/j.renene.2015.04.039
0960-1481/© 2015 Elsevier Ltd. All rights reserved.

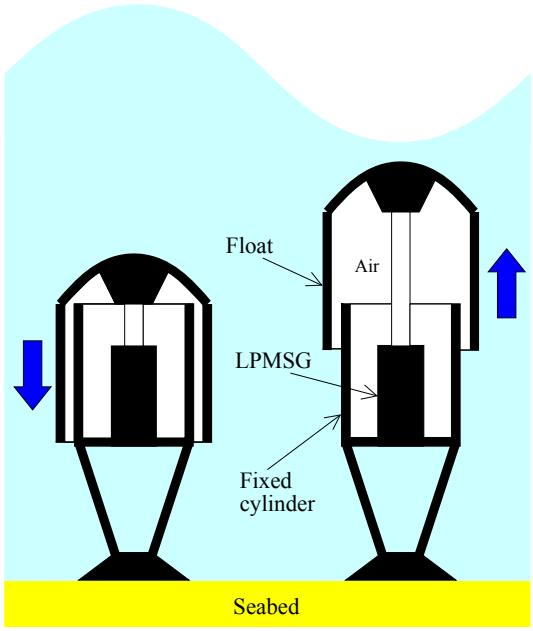


Fig. 1. The construction of the AWS.

well as the wave period (or the angular frequency of the wave). To avoid the slow response of the mechanical resonance control, a scheme based on the theory of reaction force to maximize energy extraction is discussed in Refs. [11,12]. However, the ratings of the back-to-back converter need to be increased without increase in the active power generation. Recently, the Maximum Power Point Tracking (MPPT) method of "perturb and observe" (P&O) used for solar system has been adapted to WECS [13]. The advantage of the P&O technique is its low computational burden. However, this technique is based on a time constant that needs tuning to cope with the sinusoidal time-varying nature of the generated power from WECS. Moreover, the optimal control theory is utilized to maximize the generated power of the direct-drive WECS [14]; however, its computational burden is the main disadvantage.

In an effort to overcome such drawbacks, this paper presents an MPPT technique based on speed sensorless control of the LPMSG. Eliminating the wave velocity sensor increases the reliability of the WECS. Many techniques have been proposed in the literature to estimate the velocity of the linear PM synchronous motor (LPMSM). The performance of the reduced state observer on *dq* reference frame, proposed in Ref. [15], is decayed at low and zero velocities. To overcome the drawbacks of the *dq* observer, the Model Reference Adaptive System (MRAS) is introduced in Ref. [16]. In the MRAS, some electrical quantities such as back EMF or reactive power need to be calculated. Other techniques are applied to estimate the velocity of the linear PM motors such as high frequency signal injection [17], the neural networks [18], and the Extended Kalman Filter (EKF) [19]. Although the EKF is relatively easy to apply, it is based on a linearization process which requires lengthy and complex derivation of the Jacobian/Hessian matrices and may leads to filter instability. To overcome the EKF drawbacks, the Unscented Kalman Filter (UKF) is introduced as a novel estimation tool for nonlinear filtering problems. The main advantage of the UKF is the elimination of the linearization of the system model [20,21].

In this paper, the UKF is utilized to estimate the translator velocity of the LPMSG. Moreover, the optimal translator velocity is acquired from the instantaneous active power at the generator terminals. A simple PI controller is utilized to regulate the translator velocity at the optimal value to extract the maximum power.

Grid codes for renewable energy sources require them to remain connected during grid disturbances to ride-through the faults. Moreover, this Low Voltage Ride-Through (LVRT) capability obliges the renewable energy sources to supply reactive power to assist the voltage recovery and system stability [22–25]. As a result, an adequate LVRT control must be integrated into the WECS.

This paper proposes a new control strategy for the back-to-back converter interfacing the LPMSG based AWS with the grid not only to extract the maximum power from the wave, but also to meet the LVRT requirements. The grid-side converter is used to provide the necessary reactive power to support the grid. In addition, overvoltage protection for dc-link is taken into consideration by a braking chopper along with the proposed de-excitation control technique for the generator-side converter. The performance of the proposed integrated control strategy for WECS based on LPMSG coupled to AWS is tested by using the PSCAD/EMTDC simulation package [26]. The results are discussed to demonstrate the potential of the proposed techniques.

2. Modeling of the wave energy conversion system

The direct-drive WECS, under study, consists of three main components: the AWS, the LPMSG, and the back-to-back converters with their controllers. The following subsections present the dynamic model of each component.

2.1. The AWS

The equation of motion for the AWS, shown in Fig. 1, is expressed as follows [9]:

$$F_w = F_e + m_{tot} dv/dt + \beta_h v + k_s y \tag{1}$$

where v is the linear velocity of the translator, $v=dy/dt$, y is the distance travelled by the floater and translator, m_{tot} is the total mass of the moving parts of the AWS and the added-mass of the water above the AWS that has to be accelerated, β_h is the hydrodynamic damping coefficient of the AWS, k_s is the spring constant of the AWS, F_w is the excitation force acting on the floater due to the wave, and F_e is electromagnetic force of the LPMSG.

2.2. The LPMSG

The rotor of the LPMSG, called translator, reciprocates. Therefore, the magnitude and direction of the translator velocity are changing. As a result, induced voltages and currents of the LPMSG vary in amplitude and frequency. The vectorial model of the LPMSG in *dq* reference frame fixed to the translator is given by

$$v_d = -Ri_d + L_d di_d/dt - \omega L_q i_q \tag{2}$$

$$v_q = -Ri_q + L_q di_q/dt + \omega L_d i_d + \omega \psi_{pm} \tag{3}$$

$$F_e = \frac{3}{2}\frac{2\pi}{\lambda}\left((L_d - L_q) i_d i_q - \psi_{pm} i_q\right) \tag{4}$$

where v_d and v_q are the d–q components of the stator voltage, respectively; i_d and i_q are the d–q components of the stator current, respectively; R is the stator resistance, L_d and L_q are the d–q inductances of the stator winding, respectively; ψ_{pm} is the flux linkage of the permanent magnet, λ is the pole width of the LPMSG, and ω is the angular speed of the stator variables, $\omega=2\pi v/\lambda$.

The Park transform, given by (5), is used to convert from the three-phase reference frame to the $d-q$ translator reference frame.

$$P = \frac{2}{3}\begin{bmatrix} \cos(\theta) & \cos(\theta-120) & \cos(\theta+120) \\ -\sin(\theta) & -\sin(\theta-120) & -\sin(\theta+120) \\ 0.5 & 0.5 & 0.5 \end{bmatrix}, \quad (5)$$

where $\theta = \int \omega dt - \pi/2$, since the speed leads the position of the AWS by $\pi/2$ Ref. [10].

2.3. The back-to-back converters and their controllers

The LPMSG is integrated to the grid through a back-to-back converter which consists of two current controlled Voltage Source Converters (VSC) as shown in Fig. 2. The back-to-back converter is connected to the grid at the PCC through a step up transformer *T*. The main function of the back-to-back converter is to transfer the extracted power from the LPMSG based direct-drive WECS to the power grid. The generator-side VSC is controlled to extract the maximum power from the LPMSG based AWS. The proposed maximum MPPT technique is based on speed sensorless control of the LPMSG to follow the wave. The optimal translator velocity is obtained from the measured voltages and currents at the generator terminals. Section 3 presents the details of the proposed optimal control algorithm for the generator-side VSC.

The main objective of the grid-side VSC is to deliver the power from the generator-side VSC to the grid through the dc-link. This function is simply achieved by regulating the dc-link voltage at a constant value. A PI controller is used to process the error between the measured dc-link voltage, V_{dc}, and the required voltage level, V^*_{dc}, as presented in Fig. 2. The output from the PI controller is the d-axis current command value i^*_d required for the dc-link voltage regulation. To prevent exceeding the power rating of the grid-side VSC, a limiter is designed for i^*_d as follows:

$$i^*_d|_{limiter} = 2/3\ P_g|_{rated}/V_{PCC} \quad (6)$$

where $P_g|_{rated}$ is the rated power of the LPMSG and V_{PCC} is the grid voltage.

During normal operation, the quadrature current command i^*_q is set to zero for zero reactive power injection. This is not the case during grid faults as the LVRT capability, consented to grid codes, requires the generation unit to supply reactive power to support the grid voltage to recover. Finally, the inverse Park transform is adopted to construct the reference value of the VSC three-phase currents. The phase angle θ required by the Park transform is obtained from a Phase Locked Loop (PLL) circuit.

During grid faults, the grid-side VSC fails to transmit the active power from the generator to the grid due to the deficient PCC voltage. Therefore, the dc-link voltage increases and may exceed the rated voltage of the back-to-back VSCs. Usually, a braking chopper is utilized to protect the back-to-back VSCs from over-voltage by discharging the dc-link capacitor in an external resistance R_b as shown in Fig. 2. Once the dc-link voltage reaches a predefined limit, the chopper is activated to dissipate the active power. Therefore, in such circumstances, it would be appropriate to deactivate the MPPT control of the generator-side VSC to avoid dc-link overvoltage, which is further discussed in the next section.

3. The proposed control strategy for the WECS

The maximum power is captured from the wave when the AWS is in mechanical resonance. To work at resonance, the following two conditions must be satisfied [9]:

$$m_{tot} dv/dt + K_s\, y = 0 \quad (7)$$

$$F_e = \beta_h v. \quad (8)$$

Condition (7) denotes that the resonance angular frequency of the floater, $\sqrt{K_s/m_{tot}}$, is equal to the angular frequency of the wave ω_{wave},

$$\omega_{wave} = \sqrt{K_s/m_{tot}} \quad (9)$$

To keep the resonance at different wave periods (frequencies), the spring constant, K_s, can be changed by regulating the air pressure and volume inside the AWS to satisfy (9).

Furthermore, (8) implies that the generator should be controlled in a way to keep its electromagnetic force equal to the hydrodynamic damping force. The following subsection describes the proposed control system for the generator-side VSC to achieve the MPPT.

3.1. The proposed MPPT control technique for the generator-side vsc

A block diagram of the proposed MPPT control technique for the generator-side converter is presented in Fig. 3. The generated instantaneous active power P_{gen} is calculated as follows:

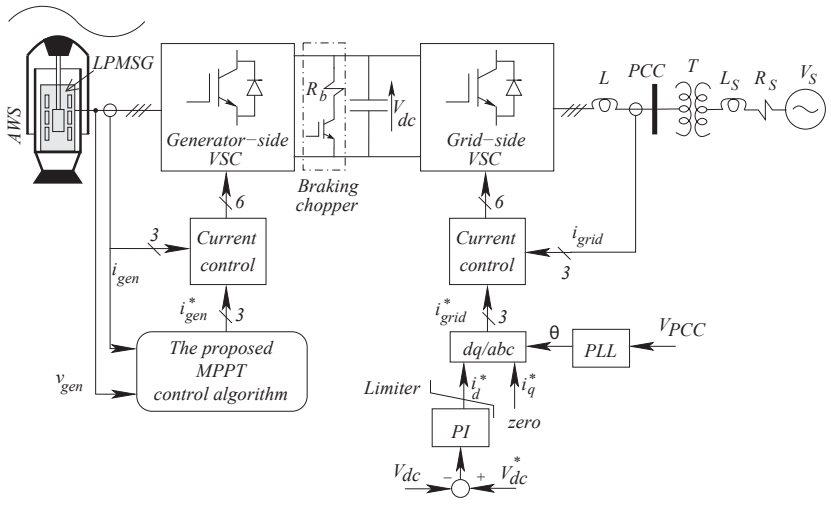


Fig. 2. Block diagram of the proposed wave energy conversion system.

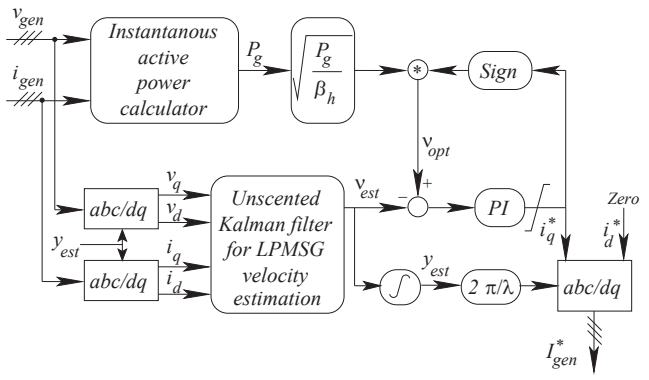


Fig. 3. Block diagram of the proposed MPPT control system for the generator-side VSC.

$$P_{gen} = v_a i_a + v_b i_b + v_c i_c \tag{10}$$

where $v_{gen}(v_a,v_b,v_c)$ and $i_{gen}(i_a,i_b,i_c)$ are the terminal phase voltages and currents of the LPMSG, respectively. A first-order Low Pass Filter (LPF) is used to filter out the high frequency ripples from P_{gen}, due to the switching action of the VSC.

Neglecting the internal losses of the LPMSG and substituting $P_g = F_e v$ in (8)

$$v_{opt} = \pm\sqrt{P_g/\beta_h} \tag{11}$$

where v_{opt} is the optimal translator velocity that results in maximum power point tracking. To regulate the translator velocity at v_{opt}, the actual value is required. Instead of measuring the translator velocity, the UKF is utilized to estimate it from the measured voltages and currents of the LPMSG, as described in the next subsection. The position of the translator required by the Park transform, *abc/dq*, is realized by a discrete integration of v_{est}.

A simple closed loop control is used to regulate the translator velocity at its optimum value v_{opt}. A PI controller processes the error between the estimated and the optimal translator velocities to generate the quadrature component of the generator-side VSC current i_q^*. The sign of i_q^* is used to determine the sign of v_{opt} in (11).

The *d*-axis command value of the stator current is set to zero, $i_d^* = 0$, to reduce the power loss in the stator windings of the LPMSG and to maximize the power-to-current operation. Finally, the inverse Park transform is employed to construct the reference value of the generator-side VSI three-phase currents. The estimated translator position y_{est} is utilized. It is worth mentioning that this action eliminates the need of using the Phase Locked Loop (PLL) circuit. Based on the Volt-Ampere (VA) rating of the generator, a limiter is designed for the active power through i_q^*.

3.2. The UKF for translator velocity estimation

In this paper, the UKF algorithm has been adapted to estimate translator velocity of the LPMSG. The UKF gives a least-square optimal estimate without linearizing the system model. The UKF algorithm is based on implementation of the unscented transformation (UT) to the nonlinear state distribution. A set of deterministic sample points around the state is selected and propagated through the nonlinear model of LPMSG. With these results a mean and covariance can be approximated using a weighted sample mean and covariance of the transformed sample points [27].

To allow the translator velocity estimation, (2) and (3) are written in the following third-order vector form, as

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{f}(\mathbf{x}(t))\mathbf{x}(t) + \mathbf{G}\mathbf{u}(t) = f(\mathbf{x}_k,\mathbf{u}_k) + \mathbf{v}_k \\ \mathbf{y}(t) &= \mathbf{H}\mathbf{x}(t) = H(\mathrm{x}_k) + \mu_k\end{aligned} \tag{12}$$

where $\mathbf{x} = [i_d \quad i_q \quad v_{est}]^T$ is the vector of state variables, $\mathbf{u} = [\mathrm{v_d} \quad \mathrm{v_q}]^T$ is the input vector, $\mathbf{y} = [i_d \quad i_q]^T$ is the output vector, superscript T is the matrix transpose operation, $\mathbf{v}_k$ and μ_k are zero mean Gaussian noise vectors, at the instant $t = k$, whose covariance matrices are **Q** and **R**, respectively. The system state matrices are expressed as follows:

$$\mathbf{f}(\mathbf{x}(t)) = \begin{bmatrix} R/L_d & 2\pi L_q v_{est}/\lambda L_d & 0 \\ -2\pi L_d v_{est}/\lambda L_q & R/L_q & -2\pi\psi_{pm}/\lambda L_q \\ 0 & 0 & 0 \end{bmatrix},$$
$$\mathbf{G} = \begin{bmatrix} 1/L_d & 0 \\ 0 & 1/L_q \\ 0 & 0 \end{bmatrix}, \text{ and } \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}. \tag{13}$$

It is noteworthy that the translator velocity of the LPMSG is assumed constant during the sampling time which results in zero derivate of its state variable, v_{est}. The mathematical details of the UKF algorithm are presented in the following three sections.

3.2.1. Sigma point selection

For the state variable **x** of dimension n with mean $\bar{\mathrm{x}}$ and covariance **P,** a set of $2n + 1$ sigma points χ_i which represents the true mean and covariance for each state x, is formulated as

$$\chi_i = \begin{cases} \bar{x} & i = 0 \\ \bar{x} + \sqrt{(n+\lambda)\mathrm{P}_{x,i}} & i = 1,\cdots,n \\ \bar{x} - \sqrt{(n+\lambda)\mathrm{P}_{x,i-n}} & i = n+1,\cdots,2n \end{cases} \tag{14}$$

where λ is a scaling factor and $\mathbf{P}_{x,i}$ is the covariance matrix element at the xth raw corresponding to a state variable x and at the ith column. The sigma points for different states are assembled in the matrix $\chi = [\chi_0 \quad \chi_1 \quad \cdots \quad \chi_{2n}]$. The weights associated with the mean $w_i^{(m)}$ and covariance $w_i^{(c)}$ elements of χ_i are given by

$$w_i^{(m)} = \begin{cases} \lambda/n+\lambda & i = 0 \\ 1/2(n+\lambda) & i = 1,\cdots,n \end{cases} \tag{15}$$

$$w_i^{(c)} = \begin{cases} w_{i=0}^{(m)} + 1 - \alpha^2 + \beta & i = 0 \\ w_i^{(m)} & i = 1,\cdots,n \end{cases} \tag{16}$$

The factors α and β express the spread of the sigma points around $\bar{\mathrm{x}}$ and prior knowledge of the distribution of $\bar{\mathrm{x}}$, respectively.

3.2.2. Time update

The UT is used to propagate the sigma points through the LPMSG model of (12) as follows:

$$\begin{aligned}\chi_k^- &= f(\chi_{k-1},\mathrm{u}_{k-1}) \\ \mathrm{Y}_k^- &= H\left(\chi_k^-\right)\end{aligned} \tag{17}$$

The weighted average means for any state variable and observation are predicted from

$$\widehat{x}_k^- = \sum_{i=0}^{2n} w_i^{(m)} \chi_{i,k}^- \tag{18}$$

$$\widehat{y}_k^- = \sum_{i=0}^{2n} w_i^{(m)} \mathrm{Y}_{i,k}^- \tag{19}$$

where $\chi_{i,k}^-$ and $Y_{i,k}^-$ are the ith rows of χ_k^- and Y_k^-, respectively. In addition, $\hat{x}_k^-$ and $\hat{y}_k^-$ corresponds to the ith elements of the predicted states vector $\hat{\mathbf{x}}_k^-$ and the observation vector $\hat{\mathbf{y}}_k^-$, respectively. Now, the error covariance matrix is predicted as follows:

$$\mathbf{P}_k^- = \left[\chi_k^- - \hat{x}_k^-\right] diag\left(w^{(c)}\right) \left[\chi_k^- - \hat{x}_k^-\right]^T + Q \tag{20}$$

where the operator $diag(w^{(c)})$ generates a zero square matrix with the elements of $w^{(c)}$ laid on the main diagonal. It is noteworthy that the term $[\chi_k^- - \hat{x}_k^-]$ is interpreted by subtracting the predicted states vector $\hat{x}_k^-$ from each column of the sigma points matrix χ_k^-.

3.2.3. Measurement update

The innovation and cross covariance matrices are calculated from

$$\mathbf{P}_{yy} = \left[Y_k^- - \hat{y}_k^-\right] diag\left(w^{(c)}\right) \left[Y_k^- - \hat{y}_k^-\right]^T + R \tag{21}$$

$$\mathbf{P}_{xy} = \left[\chi_k^- - \hat{x}_k^-\right] diag\left(w^{(c)}\right) \left[Y_k^- - \hat{y}_k^-\right]^T \tag{22}$$

Finally, the Kalman filter gain is updated and the optimal estimate and the error covariance are corrected as follows:

$$K_k = \mathbf{P}_{xy} P_{yy}^{-1} \tag{23}$$

$$\hat{\mathbf{x}}k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k \left(\mathbf{y}_k - \hat{y}_k^-\right) \tag{24}$$

$$P_k = P_k^- - K_k \mathbf{P}_{yy} K_k^T \tag{25}$$

3.3. The proposed LVRT controller

The proposed LVRT control is presented in Fig. 4. Once a fault is detected, the selector switches of i_q^*, for both VSCs, are shifted from position A for normal operation to position B for LVRT control. Setting i_q^* of the generator-side VSC to zero, during the fault, reduces the active power of the LPMSG toward zero. This action prevents dc-link overvoltage during the fault. However, at the fault instant, dc-link overvoltage may occur until de-exciting the generator-side VSC. Therefore, the brake chopper is still required to protect the back-to-back VSCs, but with a reduced energy rating. Furthermore, the grid-side VSC is controlled to inject reactive current $i_{reactive}$ to the grid according to the grid code, in contrast to the normal operation. For PCC voltages V_{PCC} lower than 0.5 pu, many grid codes require the grid-side VSC to support the grid with reactive current equal to the rated current i_{rated}. While satisfying the LVRT capability, care must be taken from the other current component of the grid-side VSC, i_d which is responsible for the dc-link voltage regulation, to prevent overcurrent trip. Accordingly, a dynamic limit $i_q^*\big|_{limit}$ is set on the reactive current acquired from the grid code to prevent the current of the grid-side VSC from exceeding its rated value. This limit is tailored as follows:

$$i_q^*\big|_{limit} = \sqrt{I_{rated}^2 - i_d^{*2}}. \tag{26}$$

Given that the currents of the grid-side VSC are tightly tracking their reference signals, it is assumed that $i_d = i_d^*$.

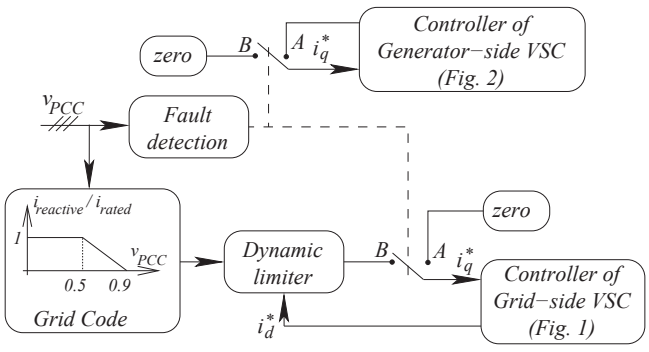


Fig. 4. The proposed LVRT Controller.

4. Simulation results

The performance of the proposed control approach for LPMSG based WECS is assessed by a computer simulation that uses PSCAD/EMTDC. The model of the LPMSG and the UKF algorithm are programmed by using C interface with FORTRAN in the EMTDC environment. The parameters of the system under study are taken from Ref. [11]. Different simulation tasks are carried out to examine the performance of the proposed WECS under various operating conditions. Primarily, the first task is devoted to examine the dynamic performance of the proposed UKF for translator velocity estimation of the LPMSG under a varying wave excitation force. The second task is dedicated to evaluate the performance of the proposed speed sensorless optimal control technique for LPMSG based WECS. Finally, the third task is assigned to assess the LVRT capability of the proposed WECS.

4.1. Dynamic performance of the UKF for velocity estimation

The sampling time for the UKF is set at 0.1 *ms*. To assess the proposed UKF algorithm for tracking the translator velocity, the period and magnitude of the wave excitation force is hypothetically changed as follows:

$$F_w = \begin{cases} 0.9 \times 10^6 \sin(2\pi t/6) & t < 24s \\ 0.9 \times 10^6 \sin(2\pi t/12) & 24 \le t < 72s \\ 0.45 \times 10^6 \sin(2\pi t/12) & t \ge 72s \end{cases}. \tag{27}$$

The actual and estimated translator velocities are depicted in Fig. 5(a), (b) illustrates the estimation error which is the difference between the actual and estimated velocities. Tight tracking with negligible error and a fast convergence are revealed. The estimation error follows a repetitive pattern. Fig. 6 represents a window from Fig. 5. The estimation error starts to increase after the zero velocity in the same direction to a limited value. This result reveals that detecting the zero crossings of the estimated velocity signal can be utilized to accurately estimate the period of the wave which is needed to keep the AWS at resonance as described in Section 3.

4.2. Performance of the proposed speed sensorless optimal control technique for the WECS

The purpose of this simulation is to evaluate the dynamic performance of the proposed optimal control technique for the WECS. Figs. 7–11 explore the dynamic behavior of the proposed direct-drive WECS. The wave excitation forceF_w is hypothetically changed with time according to (27) and is drawn in Fig. 7(b). The MPPT controller calculates the optimal reference for the translator velocity which is shown in Fig. 7(a). The generator-side VSC succeeds in regulating the translator velocity at its optimal setting, even when the excitation force changes at $t = 24$ *s* and $t = 72$ *s* with

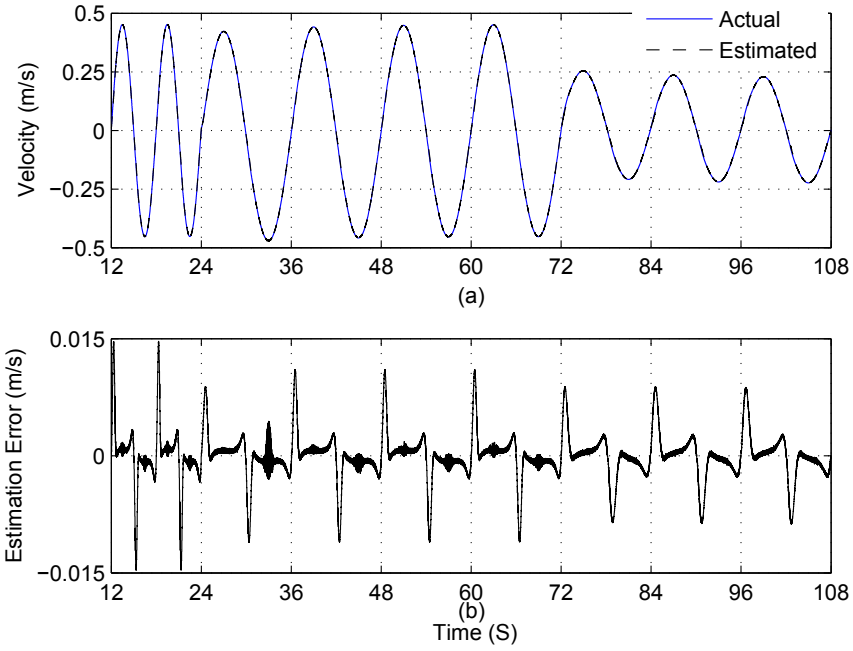


Fig. 5. Translator velocity estimation with the proposed UKF algorithm: (a) actual and estimated velocities, and (b) the estimation error.

fast dynamics and negligible overshoot. It is noteworthy that the electromagnetic force, wave excitation force, and its corresponding velocity are all in phase. This can be expressed by substituting the resonance conditions (7) and (8) in (1). Moreover, the electromagnetic force is half the value of the excitation force at any instant. This action is expected as the generator is controlled to keep its electromagnetic force equal to the hydrodynamic damping force to satisfy the resonance condition of (8). Fig. 7(c) illustrates the control action i_q^* which is responsible for regulating the translator velocity at the optimal value. This action results in extracting the maximum power from the AWS shown in Fig. 8(a). The d-axis current i_d of the generator-side VSC is regulated at zero and i_q tightly tracks its reference value i_q^*. Fig. 7(b) indicates that the electromagnetic force F_e of the LPMSG follows the waveform of i_q^*, the force producing component, as expected.

Fig. 8(a) portrays the active powers at the LPMSG and the grid terminals. The small difference between the generator and grid powers represents the losses in the back-to-back converters and the leakage impedance of the dc-link capacitor. Since the electromagnetic force and velocity changes in magnitude and direction, the active power oscillates with double the wave frequency as demonstrated in Fig. 8(a), (b) indicates the zero reactive power

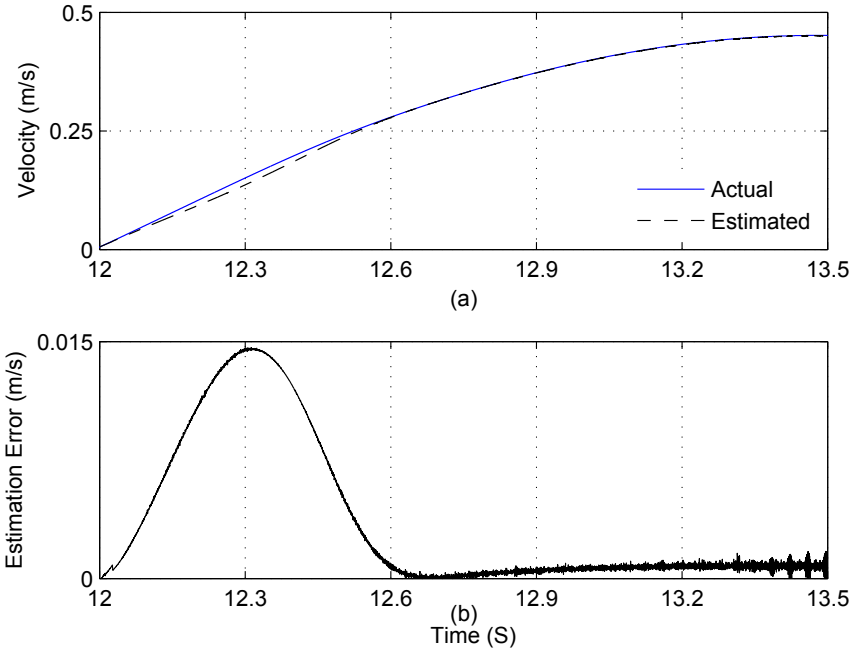


Fig. 6. Zoom from Fig. 5.

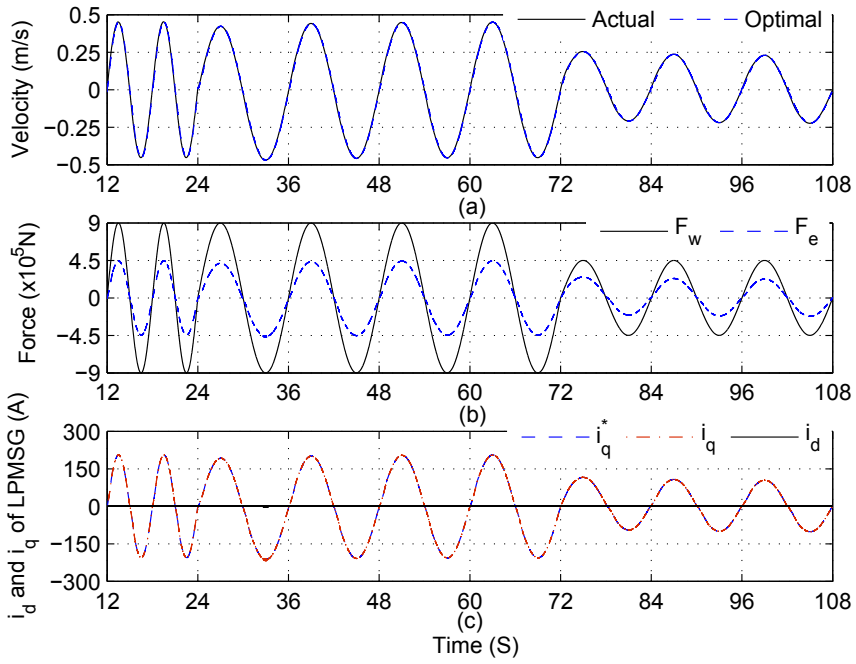


Fig. 7. Dynamics of the proposed WECS: (a) actual and optimal velocities, (b) excitation and electromagnetic forces, and (c) d–q current components of the generator-side VSC.

generation from the grid-side VSC and the oscillating reactive power delivered by the LPMSG through the generator-side VSC. The grid-side converter succeeds to regulate the dc-link voltage at its desired reference value of 1.2 kV, as illustrated in Fig. 8(c). The negligible fluctuations in the dc voltage have the same frequency of the delivered active power and are slightly increase with the increase of wave magnitude or frequency. The voltage regulation is achieved by transferring the oscillatory active power from the generator to the grid. This is achieved by controlling the d-component current command of the generator-side VSC. It is

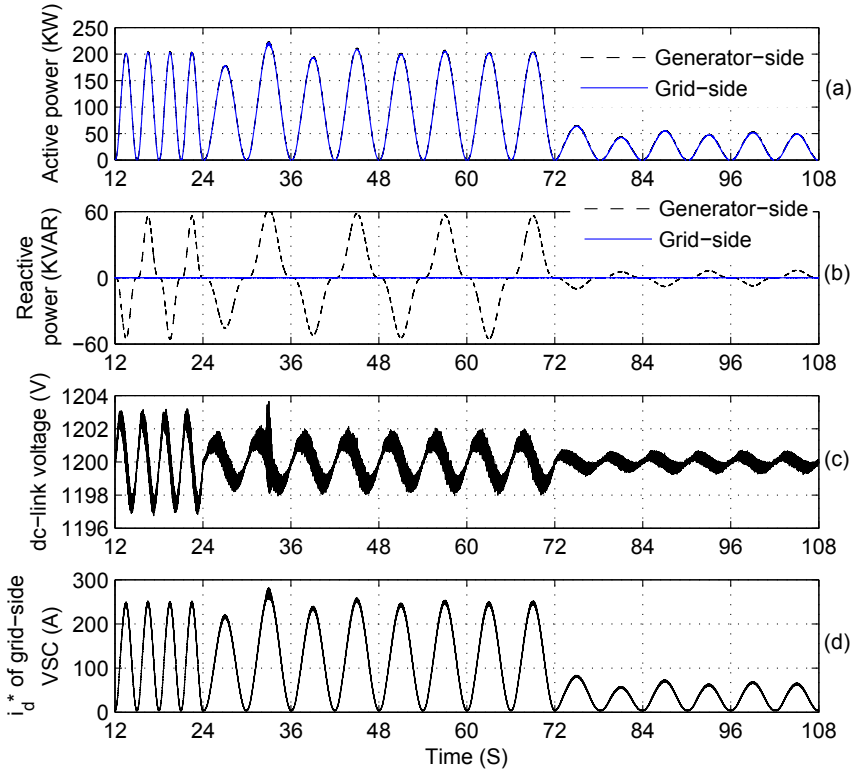


Fig. 8. Dynamic performance of the generator-side and the grid-side VSCs: (a) active power, (b) reactive power, (c) dc-link voltage, and (d) d-axis current component of the grid-side VSC.

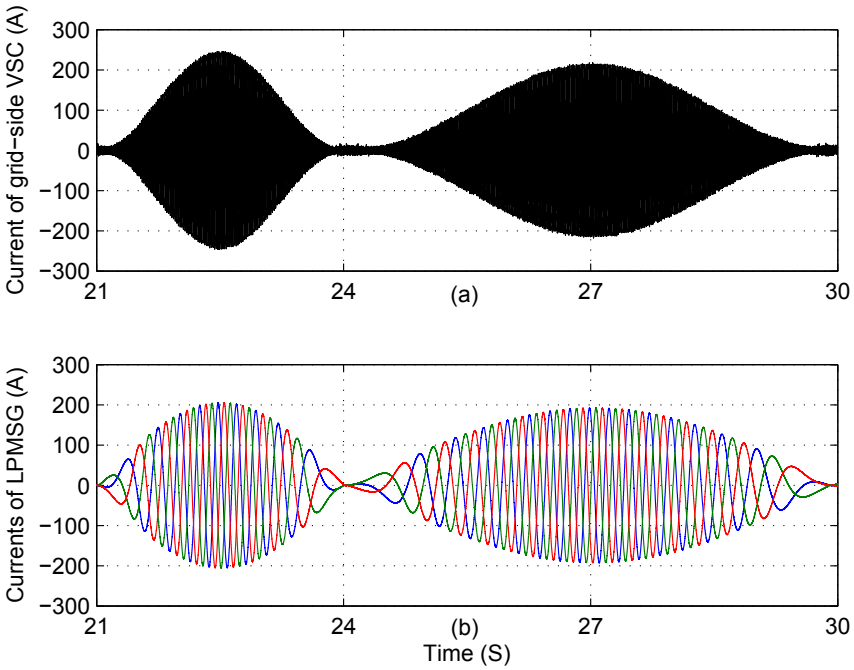


Fig. 9. Phase currents of: (a) grid-side VSC and (b) LPMSG, during the first change in the excitation force.

obvious that the active power, shown in Fig. 8(a), and the active power producing component i_d^*, shown in Fig. 8(d), exhibit similar shapes.

The phase currents of the grid-side VSC and the generator-side VSC are portrayed in Fig. 9(a) and (b), respectively, during the first change of the excitation force. It is obvious that the envelope of the grid-side VSC current is equal to its d-axis current component shown in Fig. 8(d). Similarly, the envelope of the LPMSG current is equal to the q-axis current component of the generator-side VSC displayed in Fig. 7(c).

Finally, a comparison between the performance of the proposed control strategy and the method presented in Ref. [11] is conducted. In Ref. [11], the generator-side VSC is controlled to keep the resonance conditions without regulating the spring constant K_s of the AWS. The spring constant is set to resonate at a wave period of 6s in the method of [11]. With the same parameters, Fig. 10 shows the active power generated using the two systems. For $t < 24$ s, where the wave period is 6s, the active powers generated from the two systems are identical. However, when the wave period is changed, $t > 24$, the active power generated from the two systems are different, while the energy extracted remains constant. Fig. 10 illustrates that during the transients and steady state operation, the peak value of the active power extracted using the proposed system is less than that of the method presented in Ref. [11]. Moreover, the active power extracted using the method presented in Ref. [11] becomes negative during part of the cycle. This action results from counting on the generator-side VSC to satisfy the response conditions at fixed spring constant of the AWS. This result explores the superiority of the proposed system compared with [11] in terms of reduced power fluctuations and reduced converter size.

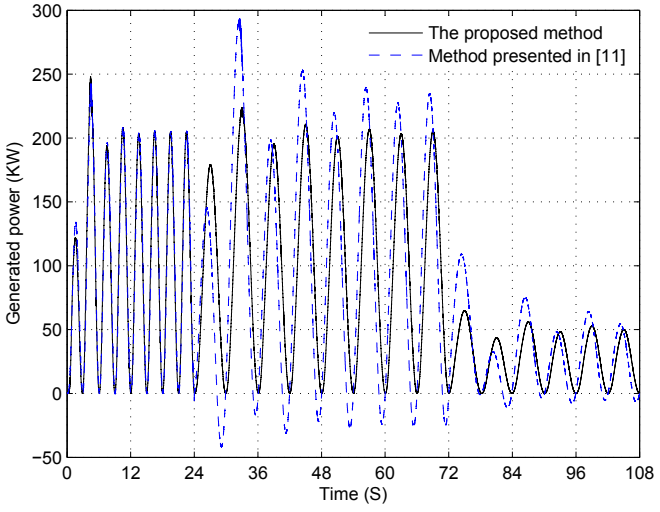


Fig. 10. Comparison between the active power generated using the proposed strategy and the method presented in Ref. [11].

4.3. Evaluation of the proposed system under irregular wave

Irregular wave can be represented by summation of sinusoids with different magnitudes and frequencies. The dominating component of the wave excitation force corresponds to the highest magnitude component of the translator velocity. To extract maximum power from the irregular wave, the AWS is controlled to resonate at the frequency of the dominating component. To evaluate the capability of the proposed MPPT technique for extracting the maximum power from an irregular wave, the following excitation force, given in Ref. [11], is considered.

$$F_w = 9 \times 10^5 \sin(2\pi\, t/12) + 4.5 \times 10^5 \sin(\pi\, t/12) \tag{28}$$

Fig. 11(a) demonstrates that the generator-side VSC succeeds in regulating the translator velocity at its optimal track which is set by the proposed MPPT technique. Fig. 11(b) illustrates the wave excitation force and the electromagnetic force of the PMSG. As in the previous case study, the electromagnetic force, the wave excitation force, and the translator velocity are in-phase which indicates that the proposed MPPT control system succeeds to satisfy the

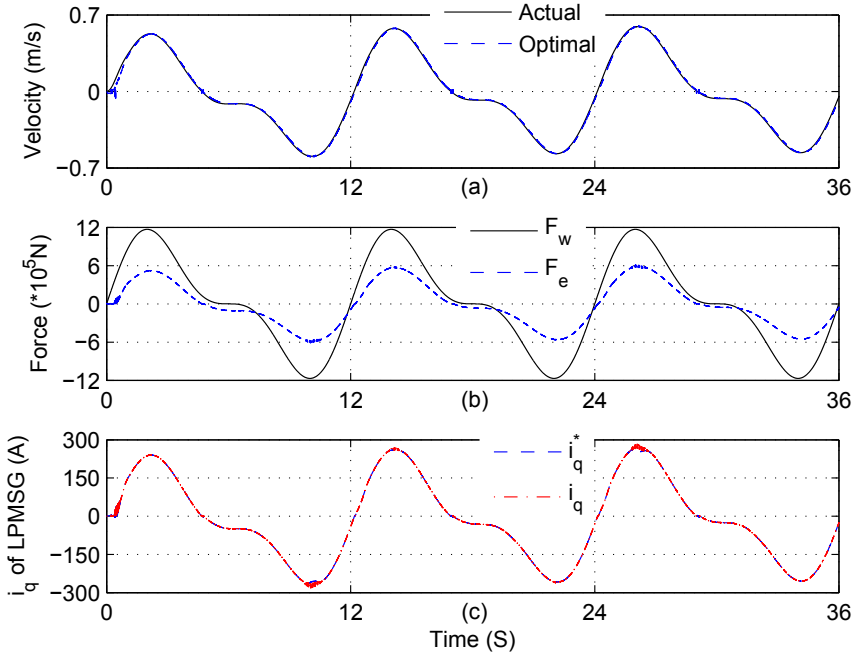


Fig. 11. Dynamics of the proposed WECS under irregular wave: (a) actual and optimal velocities, (b) excitation and electromagnetic forces, and (c) q-axis current component of the generator-side VSC.

resonance conditions under irregular wave operation. The q-axis current of the generator-side VSC is tightly tracking its reference value i_q^* as indicated in Fig. 11(c). As expected, the waveforms of i_q^* and the electromagnetic force are similar as the former regulates the translator velocity at its optimal value to extract the maximum power from the wave.

The extracted power from the wave at the generator-side and grid-side are traced in Fig. 12(a). Since the waveforms of the velocity and force have not half-wave symmetry, the active power oscillates with the frequency of the dominating component of the wave excitation force. It is clear that the active power follows its command signal i_d^*, the d-axis current reference of the grid-side VSC, as illustrated in Fig. 12(b). Since the reactive current component of the grid-side VSC is set to zero, the envelope of the grid-side VSC current, Fig. 12(c), is equal to its corresponding d-axis current command. Similarly, the envelope of the LPMSG current, Fig. 12(d), is equal to the q-axis current component of the generator-side VSC shown in Fig. 11(c).

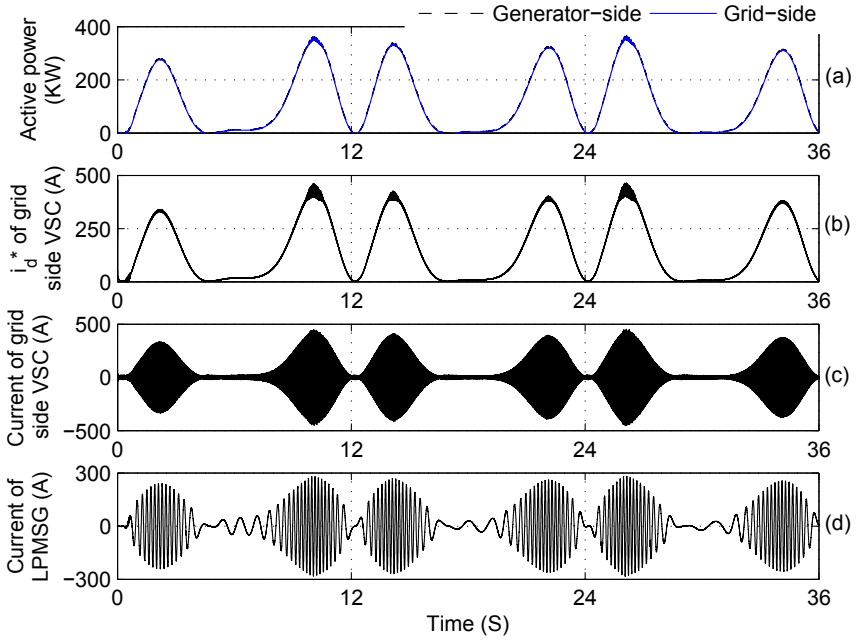


Fig. 12. Dynamic performance of the generator-side and the grid-side VSCs under irregular wave: (a) active power, (b) d-axis current component of the grid-side VSC, and phase currents of (c) grid-side VSC and (d) LPMSG.

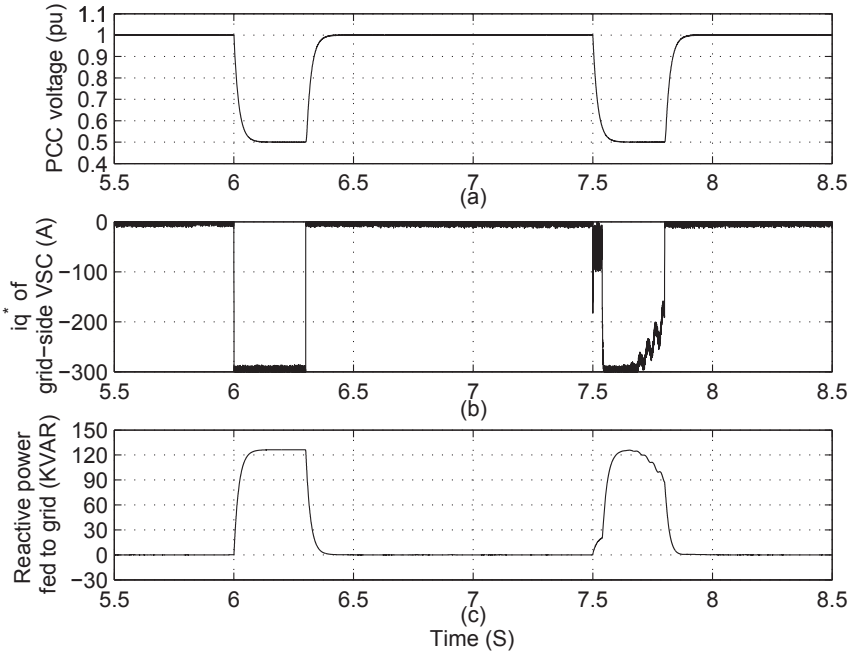


Fig. 13. LVRT capability: (a) PCC voltage, (b) q-axis current component of the grid-side VSC, and (c) injected reactive power to the grid.

4.4. Evaluation of the proposed LVRT controller

This task is dedicated to examine the LVRT capability of the proposed WECS. Two consecutive faults of 300 ms durations are introduced which results in 50% voltage dips at the PCC as indicated in Fig. 13(a). The first fault is initiated at $t = 6$ s when the generated power is minimal, while the second fault is enabled at $t = 7.5$ s when the generated active power is at peak value as shown in Fig. 14(d). Once a fault is detected, the proposed LVRT control of Fig. 4 is activated and the grid-side VSC starts to inject reactive current to satisfy the grid code, as illustrated in Fig. 13(b). For $V_{PCC} = 0.5$ *pu*, the required reactive current is equal to the rated current of the VSC, which is 300 A. The corresponding injected reactive power is portrayed in Fig. 13(c).

Furthermore, Fig. 14(c) denotes that i_q^* of the generator-side VSC is assigned to zero at the fault instant. Consequently, i_q decays to zero. However, i_q is depending on the generated power from the LPMSG at the fault instant. In the first fault, when the generated power is zero at the fault instant, i_q is successfully regulated at zero and the generator is completely de-loaded as illustrated in Fig. 14(d). As a result, the electromagnetic force becomes zero and the velocity is slightly increased as depicted in Fig. 14(b) and

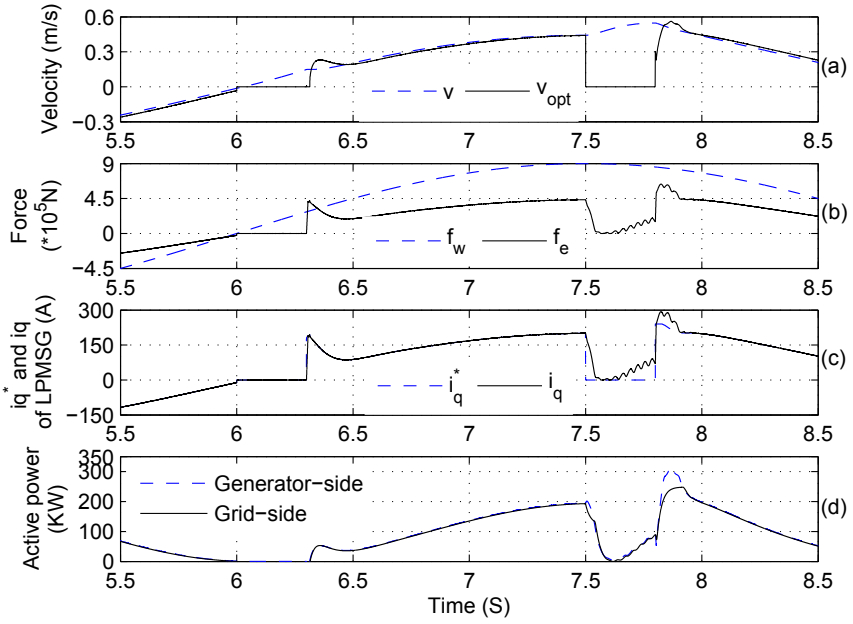


Fig. 14. Dynamics of the proposed WECS during grid faults (a) actual and optimal velocities, (b) excitation and electromagnetic forces, (c) d–q current components of the generator-side VSC, and (d) active power of the generator-side and the grid-side VSCs.

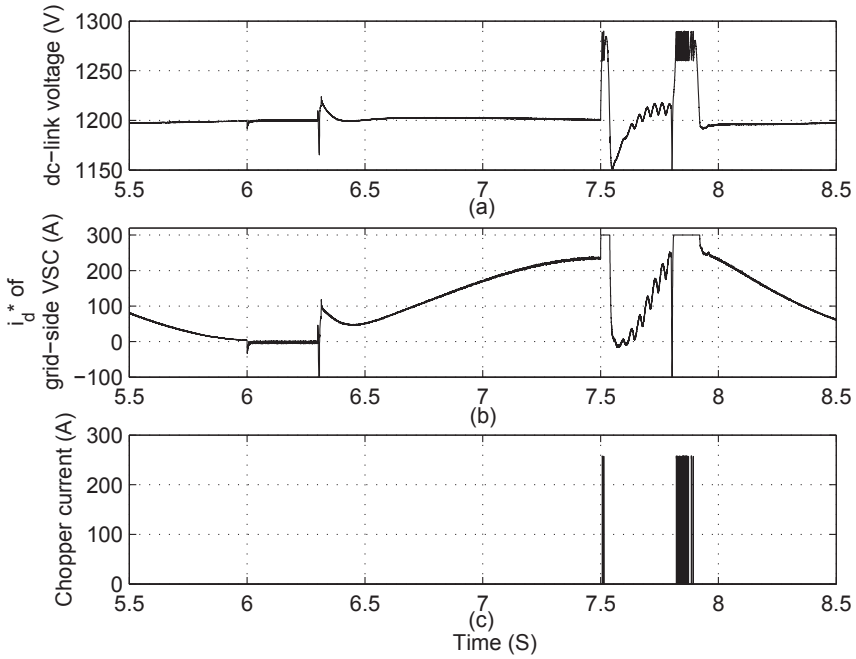


Fig. 15. Dc-link overvoltage protection: (a) dc-link voltage, (b) *d*-axis current component of the grid-side VSC, and (c) current of the braking chopper.

Fig. 14(a), respectively. On the contrary, when the generated power at the fault instant is not zero, such as the situation of the second fault, i_q cannot be maintained at zero during the fault. Once the currents of the generator-side VSC achieves zero, it works as a bridge rectifier that passes the building up power of the LPMSG to the dc-link. This is the reason why the electromagnetic force is not zero during the second fault. This power is transmitted to the grid by increasing i_d of the grid-side VSC as indicated in Fig. 15(b). The dynamic limiter adjust the reactive current command of the grid-side VSC according to (28), as signified in Fig. 13(b), to not exceed the current rating of the VSC. It is obvious that after clearing the fault, the proposed MPPT control of the WECS succeeds to recover the dynamic optimal operation.

Fig. 15(a) illustrates the successful overvoltage protection of the dc-link. The activation and deactivation voltage levels of the braking chopper are set at 1290 V and 1260 V, respectively. When the second fault is initiated, i_d^* of the grid-side VSC reaches its limit as depicted in Fig. 15(b). Hence, it is failed to regulate the rapidly increased dc-link voltage results from the active power difference between the LPMSG and the grid shown in Fig. 14(d). This momentary power difference is due to the reduction of the grid voltage. As a result, the braking chopper is instantaneously

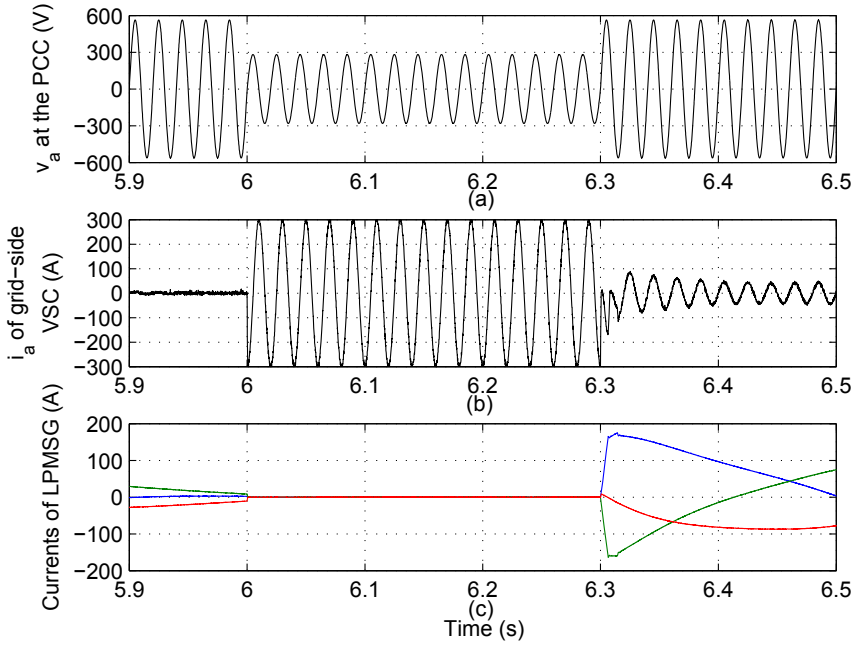


Fig. 16. The first fault: (a) phase a voltage at the PCC, (b) current of the grid-side VSC, and (c) three-phase currents of the LPMSG.

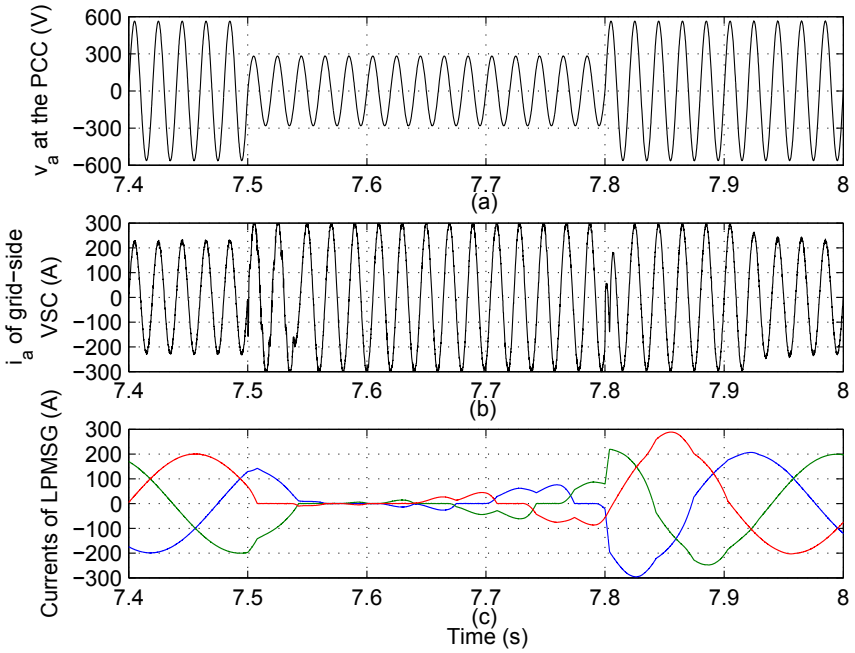


Fig. 17. The second fault: (a) phase a voltage at the PCC, (b) current of the grid-side VSC, and (c) three-phase currents of the LPMSG.

activated to limit the dc-link voltage by dissipating this power difference as displayed in Fig. 15(c). Furthermore, the braking chopper assists the dc-link voltage regulation loop of the grid-side VSC during the second fault clearance. It is worth mentioning that the proposed de-excitation control of the grid-side VSC releases the braking chopper during the fault. This action results in a reduction on the required energy rating of the braking chopper.

Fig. 16(a) portrays phase a voltage at the PCC during the first fault. Since this fault is initiated at zero active power generation, the currents of both VSCs are almost zero as indicated in Fig. 16(b) and (c). It is obvious that the phase angle between the injected current of the grid-side VSC and the PCC voltage is 90° during the fault. In addition, they become in phase after the fault clearance. Fig. 17 portrays the same signals of Fig. 16, but during the second fault. When the fault is initiated and after de-exciting the generator-side VCS, it behaves as an uncontrolled rectifier bridge and the LPMSG current starts to grow as stated before. The pulsed phase currents of the LPMSG are observed in Fig. 17(c). The results clearly show that the proposed LVRT control succeeds to fulfill the grid code requirements without exceeding the ratings of the grid-side VSC.

5. Conclusion

This paper introduces a new MPPT control technique for direct-drive WECS based on LPMSG. The proposed technique is based on speed sensorless control of the LPMSG to follow the wave. The UKF algorithm is utilized to estimate the translator velocity from the measured terminal voltages and currents of the LPMSG. The instantaneous generated power from the LPMSG is employed to obtain the optimal translator velocity. The difference between the estimated and the optimal translator velocities is passed to the controller to satisfy the resonance condition crucial to extract the maximum power. Moreover, an LVRT control is integrated to satisfy grid code requirements. Both the grid voltage and the generated active power at the fault instant are considered in dynamically limiting the injected reactive current to not exceed the ratings of the grid-side VSC. To release the braking chopper, used to protect the back-to-back converter, the generator-side VSC is de-excited during the fault. Different simulation study cases are demonstrated to assess the dynamic performance of the proposed integrated control strategy for WECS under regular and irregular waves. Tight tracking with negligible error and a fast convergence are revealed from the simulation results. The accurate estimation of the zero velocity renders the UKF suitable for detecting zero crossing of the estimated signal to precisely estimate the wave period which is needed to keep the AWS at resonance. The proposed strategy outdoes other methods in the literature, because of its direct regulation of the translator velocity and decreased power fluctuations which result in reduced converter size. Furthermore, the results clearly show that the integrated LVRT control succeeds to fulfill the grid code requirements without exceeding the ratings of the grid-side VSC.

Appendix. System Parameters

The AWS:

$m_{tot} = 0.6\text{MKg},\ \beta_h = 1.004\text{MNs/m},\ k_s = 0.65\text{MN/m}.$

The LPMSG:

$R = 0.05\Omega,\ L_d = L_q = 31\text{mH},\ \psi_{pm} = 23\text{Wb},\ \lambda = 0.1\text{m}$

The back-to-back converter:

C = 6000 μF, V_{dc} = 1200 V, L = 1 mH, rated current of the grid-side VSC = 300 A.

Step up transformer: 1.05 MVA, 690 V/22 KV, leakage reactance = 0.06 pu.

Thevenin equivalent of the grid referred to the 22 KV: V_s = 22 KV, f = 50 Hz, L_s = 6.08 mH, R_s = 0.03 Ω.

References

[1] Lopez I, Andreu J, Ceballos S, Martinez de Alegria I, Kortabarria I. Review of wave energy technologies and the necessary power-equipment. Renew Sustain Energy Rev 2013;27:413–34.

[2] O'Sullivan DL, Lewis AW. Generator selection for offshore oscillating water column wave energy converters. In: Power Electronics and Motion Control Conference, Poznan, Poland, 1–3 Sept; 2008. p. 1790–7.
[3] Kofoed JP, Frigaard P, Friis-Madsen E, Sorensen HC. Prototype testing of the wave energy converter wave dragon. Renew Energy 2006;31(2):181–9.
[4] Henderson R. Design, simulation, and testing of a novel hydraulic power take-off system for the Pelamis wave energy converter. Renew Energy 2006;31(2):271–83.
[5] Ran L, Mueller MA, Ng C, Tavner PJ, Zhao H, Baker NJ, et al. Power conversion and control for a linear direct drive permanent magnet generator for wave energy. IET Renew Power Gener 2011;5(1):1–9.
[6] Brooking PRM, Mueller MA. Power conditioning of the output from a linear vernier hybrid permanent magnet generator for use in direct drive wave energy converters. IEE Proc Gen Transm Distrib 2005;152(5):673–81.
[7] Hong Y, Waters R, Bostrom C, Eriksson M, Engstrom J, Leijon M. Review on electrical control strategies for wave energy converting systems. Renew Sustain Energy Rev 2014;31:329–42.
[8] Da Costa JS, Pinto P, Sarmento A, Gardner F. Modeling of an ocean waves power device AWS. In: Proc IEEE Conf Control Applications (CCA), Istanbul, Turkey, 25; June 2003. p. 618–23.
[9] Polinder H, Damen MEC, Gardner F. Linear PM generator system for wave energy conversion in the AWS. IEEE Trans Energy Convers 2004;19(3):583–9.
[10] Wu F, Zhang X-P, Ju P, Sterling MJH. Modeling and control of AWS-based wave energy conversion system integrated into power grid. IEEE Trans Power Syst 2008;23(3):1196–204.
[11] Wu F, Ju P, Zhang X-P, Qin C, Peng GJ, Huang H, et al. Modeling, control strategy, and power conditioning for direct-drive wave energy conversion to operate with power grid. IEEE Proc 2013;101(4):925–41.
[12] Shek JKH, Macpherson DE, Mueller MA, Xiang J. Reaction force control of a linear electrical generator for direct drive wave energy conversion. IET Renew Power Gener 2007;1(1):17–24.
[13] Amon EA, Brekken TKA, Schacher AA. Maximum power point tracking for ocean wave energy conversion. IEEE Trans Industry Appl 2012;48(3):1079–86.
[14] Abraham E, Kerrigan EC. Optimal active control and optimization of a wave energy converter. IEEE Trans Sustain Energy 2013;4(2):324–32.
[15] Wang X, Ge Q. Speed sensorless control of a linear synchronous motor using state observer on d-q reference frame. In: Proc IEEE Int Conf Electrical Machines and Systems (ICEMS), Wuhan, China, 17–20; Oct. 2008. p. 1553–7.
[16] Ng CH, Rashed M, Vas P, Stronach F, MacConnell P. A novel MRAS current-based sensorless vector controlled PMLSM drive for low speed operation. In: Proc IEEE Int Electric Machines and Drives Conf (IEMDC), Madison, WI, USA, 1–4; June 2003. p. 1889–94.
[17] Cupertino F, Giangrande P, Pellegrino G, Salvatore L. End effects in linear tubular motors and compensated position sensorless control based on pulsating voltage injection. IEEE Trans Industrial Electron 2011;58(2):494–502.
[18] Wang L, Li X. Position sensorless control for PMLSM using Elman neural network. In: Proc IEEE Int Conf information engineering and computer science (ICIECS), Wuhan, China, 19–20; Dec. 2009. p. 1–4.
[19] Chen Y-R, Cheung NC, Wu J. Sensorless drive of permanent magnet linear motors using modified Kalman filter. In: Proc IEEE Power Electronics Specialists Conf (PESC), Vancouver, BC, Canada, 17–21; June 2001. p. 2009–13.
[20] Julier SJ, Uhlmann JK. Unscented filtering and nonlinear estimation. Proc IEEE 2004;92(3):401–22.
[21] Chan T-F, Borsje P, Wang W. Application of unscented Kalman filter to sensorless permanent-magnet synchronous motor drive. In: Proc IEEE Electric Machines & Drives Conf., Miami, FL, USA; May 2009. p. 631–8.
[22] Zavadil R, Miller N, Ellis A, Muljadi E. Making connections. IEEE Power Energy Mag 2005;3(6):26–37.
[23] Molinas M, Skjervheim O, Andreasen P, Undeland T, Hals J, Moan T, et al. Power electronics as grid interface for actively controlled wave energy converters. In: Proc IEEE Int Conf Clean Electrical Power (ICCEP), Capri, Italy; May 2007. p. 188–95.
[24] Alberdi M, Amundarain M, Garrido AJ, Garrido I, Maseda FJ. Fault-ride-through capability of oscillating-water-column-based wave-power-generation plants equipped with doubly fed induction generator and airflow control. IEEE Trans Industrial Electron 2011;58(5):1501–17.
[25] Alberdi M, Amundarain M, Garrido A, Garrido I. Neural control for voltage dips ride-through of oscillating water column-based wave energy converter equipped with doubly-fed induction generator. Renew Energy 2012;48:16–26.
[26] PSCAD/EMTDC power systems simulation software user's manual. Manitoba, Canada: Manitoba HVDC Research Center, Inc.; 2003.
[27] Wan EA, Van der Merwe R. The unscented kalman filter for nonlinear estimation. In: Proc of IEEE adaptive systems for signal processing, communications, and control symposium (AS-SPCC); Oct. 2000. p. 153–8. Lake Louise, Alta., Canada.